Automatic Speech Recognition in Bengali Low-Resource Language Dialects: A Mini Review of Applications, Challenges, and Emerging Trends

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Abstract

Bangla is one of the most widely spoken languages in the world, but speech technology for its many regional dialects remains underdeveloped. Automatic Speech Recognition (ASR) for these dialects is important for real-world applications such as subtitles, voice search, and access to digital services in Bangladesh; however, most existing systems are trained only on standard Bangla. In recent years, self-supervised encoders and large multilingual models such as wav2vec 2.0, Wav2Vec-BERT, WavLM, and Whisper have facilitated building ASR systems for low-resource languages and have achieved strong results on standard Bangla. At the same time, studies on Bengali show that these models still make errors when speakers use local dialects or when recordings are noisy. This review examines the performance of different model families on Bangla, discusses how a WavLM-based framework can be adapted to handle dialect variation and noise, and analyzes insights from the Ben-10 dialect corpus regarding current system limitations. Based on these observations, the paper highlights what is already possible for Bengali dialect ASR, identifies open challenges, and outlines promising research directions for developing practical systems for Bangladeshi local dialects.

1 Introduction

Bengali is the primary language of Bangladesh and the second most widely spoken language in India, with hundreds of millions of speakers overall. From the perspective of automatic speech recognition (ASR), however, it is still treated as a low-resource language.There is much less high quality, transcribed speech available compared to languages like English or Mandarin, and most public datasets focus only on the standard form of the language. This creates a bigger issue for regional dialects, especially those spoken in rural Bangladesh, where the way people speak can be very different from the standard form Standard Colloquial Bangla (SCB). In such settings, state of the art ASR systems trained mostly on SCB may exhibit very high error rates and thus be unusable in practice.

ASR technology has advanced quickly in the last decade. Early systems like Deep Speech replaced the old methods using HMM (Hidden Markov Model) and GMM (Gaussian Mixture Model) with a deep neural network trained using a method called Connectionist Temporal Classification (CTC). This new system directly converted sound features into text and performed well on English tests, even in noisy environments [1]. However, these systems still required thousands of hours of transcribed audio, which is not practical for many languages around the world.

Two recent advances are especially useful for low resource languages. First, self-supervised learning (SSL) allows us to learn strong speech models from large amounts of unlabelled audio. For example, wav2vec 2.0 learns speech patterns and then uses a Transformer model to improve accuracy. When this model is fine-tuned with a small amount of transcribed data, it performs very well, even with just a few minutes of audio [2]. WavLM takes this further by adding noise reduction training and using a larger, more varied dataset, making the model more effective in noisy environments and for different speech tasks [3]. Second, large multilingual foundation ASR models have become popular. Whisper is a key example: its trained on around 680,000 hours of audio and text in multiple languages, and it can handle tasks like speech recognition, translation, and language identification without needing specific training for each task [4].

These models are useful for low resource languages because they can learn from unlabelled audio or transfer knowledge from other languages, reducing the need for costly manual transcription. However, it's unclear whether they can work well for Bengali regional dialects. Most research on Bangla ASR has focused on standard varieties, while dialects are often overlooked or only addressed as a minor issue. Recent studies are beginning to question this. First, Ridoy *et al.* compare Whisper and Wav2Vec-BERT on Bangla ASR using only publicly available data and show that Wav2Vec-BERT outperforms Whisper in accuracy and efficiency, especially with limited resources [5]. Second, Biswas *et al.* propose a WavLM-based framework designed for noisy, dialectal Bengali, which shows significant improvements over wav2vec-2.0 and Whisper [6]. Third, Tashwar *et al.* introduce the Ben-10 corpus, a 78-hour dataset covering ten regional dialects in Bangladesh, and demonstrate that even advanced models struggle with these dialects, both before and after fine tuning [7].

In this review we focus on three key contributions and the models behind them. Instead of covering all research on Bangla ASR, our goal is to provide a brief overview of applications and emerging trends in ASR for low resource Bengali dialects, specifically for Bangladeshi varieties. The rest of the paper is structured as follows: Section 2 reviews the current state of Bengali ASR for low-resource dialects, with a focus on the comparison between Whisper and Wav2Vec-BERT, the WavLM-based framework for dialectal speech, and the Ben-10 corpus. Table 1 at the end of Section 2 summarizes the main papers and datasets used in this review. Section 3 discusses the applications and emerging trends highlighted by these studies and uses Table 2 to summarize common problems and solutions. Finally, Section 4 concludes with a look at open problems and future research directions for Bangladeshi dialect ASR.

2 State of the Art in Bengali ASR for Low-Resource Dialects

Research on Bengali dialect ASR is influenced by three key developments: self-supervised acoustic models, large multilingual models, and dialect-specific datasets. This section focuses on three recent studies about Bengali ASR and explains their models.

Ridoy *et al.* explore how well ASR models adapt to low resource Bangla by comparing Whisper and Wav2Vec-BERT [5]. They use the Bangla subset from Mozilla Common Voice 17 and OpenSLR SLR63, which together provide about 90-100 hours of standard Bangla speech. The authors fine-tune Whisper (both Small and Large-V2) and Wav2Vec-BERT, testing with different dataset sizes and training durations. The main finding is that Wav2Vec-BERT performs better, with lower word error rate (WER) and character error rate (CER) than Whisper while using fewer resources. For example, with 70,000 training samples and eight training rounds, Wav2Vec-BERT achieves a WER of about 14.4% and a CER of 2.7%, while Whisper's error rates are higher. The study shows that Wav2Vec-BERT is more efficient for low-resource training, making it a good choice for adapting to Bengali dialects.

While Ridoy *et al.* focus on standard Bangla, Biswas *et al.* address the challenges of dialectal and noisy Bengali speech with a framework for improving ASR [6]. They point out that Bengali ASR faces two main problems: the wide variety of dialects and frequent background noise in real-world recordings, such as traffic or crowded markets. Although models like wav2vec-2.0 and Whisper are powerful, they are not specifically trained to handle these challenges. Biswas *et al.* propose a solution using WavLM, a self-supervised model that was trained with a denoising task on a large and diverse English speech dataset [3]. WavLM's training involves adding noise and echoes to the audio and teaching the model to clean it up. This approach makes WavLM more robust to noisy and distorted recordings.

The framework uses a two step fine tuning process. In the first step, WavLM-Large is adapted to standard Bengali using the Common Voice 17 Bangla and OpenSLR SLR63 datasets, providing a solid linguistic foundation. In the second step, the model is fine-tuned on dialectal Bengali speech with added noise. Data from various dialects, like Chittagonian and Sylheti, is combined with noise from the MUSAN dataset, simulating different levels of background noise. The authors test the model's performance using word error rate (WER) across different dialects and noise levels, comparing it to two strong models: a fine-tuned wav2vec-2.0 model and a Whisper-based system. The WavLM-based model outperforms both, with the biggest improvements seen at low signal-to-noise ratios (SNR). For instance, at 0 dB SNR, WavLM reduces WER by over 40% compared to wav2vec 2.0 and nearly 50% compared to Whisper [6]. These results show that combining noise aware pre-training and dialect-specific fine tuning significantly improves Bengali dialect ASR.

Tashwar *et al.* approach the issue from a different angle by creating a dedicated dialect corpus and benchmark [7]. Their Ben-10 dataset includes over 78 hours of speech-to-text data from 394 speakers across ten regions of Bangladesh, with each region contributing more than five hours of audio. The 16,690 clips (averaging about 16.6 seconds each) cover 155 topics and focus on spontaneous, everyday conversations recorded in natural, often noisy settings. Analysis of the acoustic features shows clear differences between the Ben-10 dataset and standard Bengali speech datasets, confirming that it is a unique, OOD(out of distribution) resource.

Using the Ben-10 dataset, the authors test several ASR systems: wav2vec-2.0 fine-tuned on standard Bengali (SCB), the same model fine-tuned on Ben-10, a large Conformer-based model called Hishab trained on around 20,000 hours of pseudo-labelled Bengali audio, Whisper-large-v3, Google's commercial ASR, and other competition-winning systems. In a zero-shot setting, all systems perform poorly on Ben-10, with high word error rate (WER) and character error rate (CER) across all dialects. Fine-tuning on Ben-10 improves results but still doesn't match the performance seen on standard Bengali, and significant differences between dialects remain [7]. The authors conclude that current foundation models are not yet advanced enough to handle regional dialects in low resource languages, and that dedicated dialect-specific corpora and training are essential.

Together, these three studies highlight the current state of Bengali low resource dialect ASR. Wav2Vec-BERT is shown to be more efficient in terms of data and computational resources compared to Whisper for standard Bangla [5]. WavLM, with its denoising pre-training and multi stage fine tuning, provides better performance for dialectal and noisy speech [6, 3]. Ben-10, as a challenging benchmark, demonstrates that even large, well trained models still struggle with regional Bangladeshi dialects, particularly in natural, spontaneous conversations [7]

To give a compact overview of the literature and resources discussed above, Table 1 summarises the main reviewed papers and the related datasets.

Table 1: Overview of Key Studies, Their Contributions and Associated Datasets for Automatic Speech Recognition System

Paper	Main focus	Data / languages	Main contribution and relevance	Dataset / resource	What it covers	Size	Notes for Bangladesh dialect work
Hannun et al. (2014) [1]	End-to-end ASR (Deep Speech)	English (LibriSpeech, Switchboard)	Introduces Deep Speech, a large scale end-to-end RNN + CTC ASR system. Important as an early end-to-end baseline and motivation for moving away from traditional GMM and HMM pipelines towards neural ASR that can later be adapted to low-resource languages.	TIMIT dataset	English read & conversational speech	LibriSpeech 1000 hours	purely English; no Bangla/dialect coverage
Baevski et al. (2020) [2]	Self-supervised ASR (wav2vec 2.0)	Mainly English (LibriSpeech)	Proposes wav2vec 2.0, a self-supervised framework for learning speech representations from raw audio. Shows that strong ASR can be trained with only small amounts of labelled data. Provides the main SSL backbone concept used in later Bangla and dialectal work.	LibriSpeech (mainly) & other English audio	Self-supervised English speech representations	Pretraining 60,000 hours	Adds lexical diversity but no Bangla pretraining mentioned; later works adapted it to Bangla
Chen et al. (2022) [3]	Noise-robust SSL (WavLM)	English (large multi-corpus mix)	Introduces WavLM, which uses masked speech denoising pre-training for noise-robust speech representations. Forms the technical basis for the Bengali dialect ASR framework that targets noisy real-world recordings.	Large multi-corpus mix (English-heavy, some multilingual)	Noise-robust representations; basis for Bengali dialect work	pretraining 60k-100k hrs mix & Not specified for Bangla	Includes dialects in later Bengali adaptations; technical backbone for noisy real-world Bangla ASR
Radford et al. (2023) [4]	Foundation ASR (Whisper)	Multilingual, many tasks	Presents Whisper, a large-scale multilingual and multitask encoder–decoder ASR model trained on web-scale weak supervision. Serves as a strong baseline for Bangla and as a foundation model whose limitations on dialectal Bengali are highlighted by later work.	Web-scale weakly supervised multilingual	Multilingual ASR , translation , many tasks	680,000 hours total multilingual	Strong baseline for Bangla, but limited dialect diversity
Ridoy et al. (2025) [5]	Bangla ASR model comparison	Bangla (Common Voice v17, OpenSLR SLR63)	Compares Whisper and Wav2Vec-BERT for Bangla ASR. Shows that Wav2Vec-BERT can outperform Whisper in both accuracy and efficiency under low resource conditions. Motivates using self-supervised encoders as backbones for Bengali dialect ASR.	Common Voice Bangla v17; OpenSLR Bangla (SLR63)	Mostly standard Bangla read speech; Bangla read speech with many prompts	∼54 hrs; ∼40 hrs	Good for base adaptation and speaker diversity, but has limited explicit dialect coverage. Adds lexical and prompt diversity; still focused on standard forms.
Biswas et al. (2025) [6]	Bengali dialect ASR with noise	Bengali (standard + dialectal OOD speech)	Proposes a unified denoising and adaptation framework based on WavLM with multi stage fine tuning: first on standard Bangla, then on dialectal and noisy speech. Evaluates on dialects such as Chittagonian and Sylheti under controlled noise levels and shows clear gains over wav2vec 2.0 and Whisper.	OOD-Speech (dialect set)	Challenging out-of-distribution Bangla dialect speech	∼20 hrs (reported)	Used for dialect specialisation and evaluation in the WavLM framework. Includes dialects such as Chittagonian and Sylheti which are helpful for fine-tuning dialect-aware models.

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Paper	Main focus	Data / languages	Main contribution and relevance	Dataset / resource	What it covers	Size	Notes for Bangladesh dialect work
Tashwar Dipto et al. (2025) [7]	Dialect corpus and benchmarking (Ben-10)	Bangladeshi regional dialects	Introduces Ben-10 (over 78 hours, 10 regions, 394 speakers) and benchmarks several ASR foundation models. Shows that these models struggle on dialect speech and that dialect specific training is needed. Provides a key out-of-distribution testbed for Bangladeshi dialect ASR.	Ben-10	10 Bangladesh regions, mostly spontaneous speech	78+ hrs; 16,690 clips	Dialect-focused benchmark with strong regional diversity. Shows a large OOD shift compared to standard Bangla and is essential for testing Bangladeshi dialect ASR.
de Zuazo et al. (2025) [8]	Whisper + LM integration (Whisper-LM)	Multiple low-resource languages	Shows that integrating external language models with fine tuned Whisper improves ASR performance in low resource languages. Explores both n-gram and neural LMs. Relevant for improving decoding of rare and dialect specific words in Bengali ASR.	—	—	—	—
Li & Niehues (2025) [9]	Contextual learning and LM integration	Endangered low-resource languages (5)	Studies tokenization choices and language-model integration in ASR for endangered languages. Finds that fine grained tokenization and carefully tuned LM integration can improve performance. Provides guidance for LM use in Bengali dialect ASR.	—	—	—	—
Dhahbi et al. (2025) [10]	Synthetic speech and augmentation for low-resource ASR	Bengali, Hindi, Turkish, Urdu	Proposes an end-to-end ASR system that uses FastSpeech based synthetic speech and spectrogram augmentation. Shows that joint training on real, synthetic and augmented data reduces WER/CER, including for Bengali. Informs data augmentation strategies for Bangladeshi dialects.	IARPA Babel subsets (used)	Conversational telephone speech	Bengali $\sim$62 hrs; Turkish $\sim$10 hrs (reported)	Shows that conversational telephone speech differs from read speech and can be used to simulate more realistic call centre style conditions for Bengali ASR.
Hafiz Imam et al. (2025) [11]	Low-resource ASR survey (African)	African low-resource languages	Surveys challenges and future directions for African low-resource ASR: data scarcity, dialect diversity, SSL and multilingual transfer, and ethical issues. Provides a broader context and fairness perspective that is relevant when designing inclusive Bengali dialect ASR.	—	—	—	—
Yang et al. (2025) [12]	Large-scale corpus pipeline (GigaSpeech 2)	Thai, Indonesian, Vietnamese	Presents GigaSpeech 2 (about 30,000 hours) and an automated crawling, transcription and refinement pipeline using Whisper and forced alignment. Offers a scalable recipe that can inspire semiautomatic collection of Bangladeshi dialect speech.	GigaSpeech 2	YouTube audio-only corpus (multi-domain)	$\sim$30,000 hrs	Not Bengali, but demonstrates an automated pipeline for large scale crawling, transcription and refinement in low resource languages. Can inspire similar pipelines for Bangladeshi dialect audio from online platforms.

3 Applications and Emerging Trends

Although the reviewed work is recent, it already highlights some practical trends for ASR in Bangladeshi local dialects.

One trend is the choice of models in low-resource settings. Research by Ridoy *et al.* shows that when training data and computational resources are limited, self-supervised encoders like Wav2Vec-BERT can outperform larger models like Whisper for Bangla [5]. Combining this with the WavLM results, it suggests a design approach where a single self-supervised model is first adapted to standard Bangla and then specialized for dialects. In real world systems in Bangladesh, such as voice based services or local call center automation, this design may be more efficient than using larger Whisper models, which require more resources at inference time and may be harder to adapt to dialects

A second trend is improving noise robustness for real world environments. Biswas *et al.* focus on situations where speakers use ASR in noisy places like streets, buses, or markets—common conditions in both Bangladeshi cities and rural areas. Their experiments show that models not trained to handle noise (such as standard wav2vec-2.0 or Whisper fine tuned only on clean data) perform poorly in noisy conditions. In contrast, WavLM, which is fine tuned with simulated noise, shows much better performance with lower word error rates (WER) [6, 3]. These findings highlight the importance of noise aware training for applications like automatic captioning for local news, voice search in mobile apps, or transcribing phone calls. For such tasks, robust performance requires both noise-aware pre-training and specific noise augmentation during model adaptation.

A third emerging trend is the growing importance of dialectal corpora as essential resources, not just for evaluation. The Ben-10 corpus goes beyond being a test set; it is designed for both training models and conducting detailed linguistic analysis across regions [7]. For Bangladeshi ASR systems, this suggests that community driven data collection recording speakers from different areas with information about their region, age, and gender is crucial. Without such resources, issues specific to regions like Chattogram or Sylhet will be hidden behind overall word error rates (WER) that mainly reflect Standard Colloquial Bangla (SCB).

Fourth, these studies suggest that multi stage and domain specific fine tuning can be a useful strategy. Instead of training one model on a mixed dataset, the WavLM framework splits adaptation into two stages: one for standard Bangla and one for dialects and noise [6]. This method helps the model first learn the basic phonetic and writing patterns of standard Bangla and then specialize in dialects and noise conditions. For Bangladeshi local dialects, this multi stage approach could be expanded, for example, by adding a stage for code switching between Bangla and English in urban speech, or for specific sectors like healthcare and agriculture. Research on low resource and endangered language ASR suggests that language model integration and tokenization should be customized for Bengali dialects, rather than using settings from high resource languages [8, 9, 10, 11].

A fifth trend highlights the limitations of zero-shot generalization in current foundation models. While Whisper is often marketed as a robust, ready-to-use system, the Ben-10 results show that its performance on regional Bangladeshi dialects in a zero-shot setting is far from ideal [7, 4]. Similarly, large models like Hishab Conformer, which are trained on thousands of hours of pseudo labelled Bengali audio, still struggle to adapt to dialectal speech. This suggests that foundation models cannot fully replace the need for dialect specific data and adaptation, at least with the current technology.

Finally, there is an important fairness aspect to consider. Although the Bengali studies don't always discuss fairness directly, the performance differences across dialects in Ben-10 and the significant improvements seen with dialect focused training suggest that current systems may leave out speakers from under represented regions [7, 11]. For Bangladeshi applications, this could have serious social consequences: a voice-based government service that works well for Dhaka centric speech but struggles with rural dialects could worsen existing inequalities. Therefore, including fairness aware evaluations such as reporting word error rates (WER) for each dialect and demographic group should become a key part of future Bengali ASR research.

Table 2: Challenges in Bangla Dialect ASR and Proposed Solutions from the Reviewed Studies

Problem pattern in Bangladesh dialect ASR	Evidence from reviewed papers	Solution directions suggested by the papers
Dialect speech is out-of-distribution for standard trained ASR	Ben-10 shows a strong distribution shift and poor performance of foundation models and other ASR systems on dialect speech, even after fine-tuning on standard Bangla [7].	Build dialect corpora and train dialect aware models. Use dialect specific training and evaluation splits and report metrics per dialect rather than a single global WER [7].
Noise and channel mismatch	Bengali dialect ASR is harmed by real world noise and channel conditions. The WavLM-based framework evaluates performance down to 0 dB SNR and shows large degradation for models not trained with noise [6].	Use noise-robust self-supervised backbones (such as WavLM) and apply noise augmentation during dialect adaptation. Treat noisy and reverberant conditions as a core part of training rather than an afterthought [6, 3].
Compute limits and scaling limits	Whisper can require high compute, while Wav2Vec-BERT can be more efficient. The Bangla comparison study also observes diminishing returns when only adding more data and epochs without careful tuning [5].	Pick a model that matches available hardware. Tune fine tuning length and batch sizes carefully. Focus on better data, better pre-training and better decoding strategies instead of only scaling model size [5, 2].
Weak decoding for rare and dialect words	Low-resource settings have many rare words and dialect forms. Work on endangered languages and Whisper-LM shows that tokenization and language-model integration strongly affect final ASR quality [9, 8].	Use fine-grained tokenization (e.g., characters or small subwords) and integrate external language models (n-gram or neural) into decoding. Tune LM weights and penalties for Bengali specifically rather than copying settings from other languages [9, 8].
Unequal performance across speaker groups and regions	Surveys on low resource languages report that ASR systems often perform worse for certain regions and demographic groups, and that dialects are systematically under served [11]. Results on Ben-10 also show large differences between dialects [7].	Evaluate models by dialect and by demographic subgroup where possible. Treat inclusion and fairness as design requirements. Collect additional data for under represented dialects and consider reweighting or targeted adaptation to reduce gaps [11, 7].

4 Conclusion and Future Work

The recent literature on Bengali low-resource dialect ASR paints a nuanced picture. On the one hand, advances in self-supervised learning and weakly supervised foundation models have dramatically improved the feasibility of building ASR systems for Bangla with limited labelled data. wav2vec 2.0 and its successors show that high-quality acoustic representations can be learned from large unlabelled corpora and adapted efficiently; Wav2Vec-BERT, in particular, has been shown to outperform Whisper on standard Bangla under realistic resource constraints [2, 5]. On the other hand, the introduction of Ben-10 and the WavLM-based dialectal framework demonstrate that regional dialects and noisy conditions remain open challenges. Foundation ASR models, even when large and carefully trained, do not generalise reliably to Bangladeshi local dialects in zero-shot mode, and dialect-specific fine-tuning, while helpful, does not fully close the gap [6, 7].

Additionally, creating a new dataset from local Bangladeshi regions such as Mymensingh, Noakhali, Sylhet, and Rangpur could significantly improve model performance. By collecting content from YouTube and Facebook bloggers, it is possible to capture Bangla-English mixed speech, reflecting the real hybrid nature of modern Bengali, especially with new words emerging among Gen Z speakers. This dataset would complement existing corpora like Ben-10, which mainly use standard Bangla, allowing models to better adapt to regional variations and contemporary usage. Such a dataset could serve both for training and comparative evaluation, ensuring that ASR systems remain relevant for current, real-world speech patterns.

Looking ahead, several directions appear particularly promising for research focused on automatic speech recognition in Bengali low-resource dialects, especially in the Bangladeshi context. First, expanding and refining dialectal corpora is crucial. Ben-10 provides an excellent starting point, but additional data from under-represented districts and social groups, collected with consistent transcription and annotation protocols, would enable more robust training and evaluation [7]. Even a smaller, well-curated corpus of Bangladeshi local dialects can be highly valuable if it is designed with clear coverage goals and reliable metadata.

Second, systematic comparison and combination of model families remains important. Building on Ridoy *et al.* and Biswas *et al.*, one can investigate hybrid approaches that use Wav2Vec-BERT or WavLM encoders but borrow decoding strategies from Whisper-style models, potentially augmented with external language models trained on dialect-rich text [8, 3, 5, 6]. Parameter-efficient fine-tuning methods may facilitate maintaining both a shared core and dialect-specific variants without excessive compute.

Third, deploying these systems in realistic Bangladeshi applications will require continued attention to noise robustness and code-switching. The WavLM results show that noise-aware training substantially improves performance at low SNR; extending this work to include Bangla–English code-switched speech and telephone-channel distortions would bring the models closer to real deployment scenarios such as call centres or mobile voice interfaces [6, 10]. Fourth, future evaluations should incorporate fairness-oriented metrics, reporting not only overall WER but also stratified scores by dialect, gender, age and region. Aligning Bengali dialect ASR research with broader work on low-resource and inclusive ASR would help ensure that technological advances benefit speakers across Bangladesh rather than amplifying existing imbalances [11].

In summary, the current state of research suggests that Bengali dialect ASR is technologically feasible but far from solved. The combination of self-supervised encoders like Wav2Vec-BERT and WavLM, dialect-aware multi-stage adaptation strategies and dedicated resources such as Ben-10 and automatically crawled corpora provides a solid foundation. Building on this foundation with Bangladeshi local dialect data, careful model evaluation and fairness-aware design will be key to realising practical, inclusive ASR applications for Bengali speakers in diverse regions and social contexts.

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